Methodology Passport Card

Trust in AI-Generated Medical Recommendations Among Patients

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GitHub destination (public availability not verified): https://github.com/marzhanolzhabek/ai-medical-trust-research

1. Research type

Quantitative, observational, cross-sectional survey (planned). Adult patients are divided into three observed age cohorts: 18–30, 31–50 and 51+. All participants answer the same 1–5 trust question once. No random assignment and no causal inference.

2. Research questions

RQ1: Does trust differ among the three patient age groups? RQ2: Are AI knowledge, perceived accuracy, explanation clarity and doctor supervision associated with trust? RQ3: Is prior AI use associated with trust after age-group adjustment?

3. Testable hypotheses

Primary H0: mean trust is equal in all three age groups. Primary H1: at least one group mean differs. RQ2 H0: all adjusted associations are zero; H1: at least one is nonzero. RQ3 H0: prior AI use has no adjusted association; H1: prior AI use is positively associated with trust.

4. Variables

Independent (primary): observed patient age cohort 18–30, 31–50, 51+. Independent (secondary): prior AI use yes/no and four 1–5 ratings: AI knowledge, accuracy, clarity, doctor supervision. Dependent: self-reported trust score 1–5. Controlled: common questionnaire, score anchors, age inclusion threshold and validation rules; fixed input for software smoke test.

5. Metrics and baseline

Primary: mean trust per age cohort and differences between means, with 95% confidence intervals for a future real-data study. Secondary: median, standard deviation, sample size and adjusted associations. Guardrails: valid completion >=90%, invalid/duplicate records <=10%, no identifiable patient data. Baseline/reference: 18–30 age group, not a randomized control group or competing algorithm.

6. Executable pilot

Python 3.11.9 standard-library script reads the 9-row synthetic CSV, validates records, assigns age cohorts, calculates group means and writes results/test_run.json. Config and Dockerfile are included; test suite checks group boundaries. Runtime in ms is a pipeline diagnostic, not a trust metric.

7. Research ethics and limitations

Nine records are simulated, three per cohort; they are NOT real patient responses. Do not use them to accept or reject hypotheses. Obtain informed consent, anonymize real responses and seek institutional ethics approval where required. Plan adequate sample size before inferential tests.